

Did you know?

- E-commerce giant Amazon introduced 3D printing to deliver customised products like bracelets and earrings to its customers within the shortest possible time
- Major aerospace companies like Airbus and Boeing are making significant investments in metal 3D printing
- Hindustan Aeronautics Limited has started 3D printing of aircraft engine components
- Shoe manufacturer, Nike, has been able to speed up its product development phase using 3D printing
- 3D printers are available in market that can print printed circuit boards (PCBs). These PCBs can be made of the same size and thickness as a standard 1.6mm (1/16-inch) thick, through-hole soldered perboard. The components can be fairly easily connected together without using solder. This technique is somewhat faster than the standard method of etched and soldered circuit boards. But it does require some skill and attention to detail to create a reliable circuit.
- 3D printing has helped Volvo reduce the production time of a number of its manufacturing tools by 94 per cent.

sectors has opened up opportunities to introduce more product lines, enabling rapid prototyping and fast manufacturing processes, especially for products and components with small production volumes.

3D printers offer huge cost and time reductions in certain industrial applications, while providing even better accuracy as compared to conventional manufacturing processes. This, in turn, helps SMEs and start-ups to reduce their cost of production and enables large companies to make products cheaper, faster and better. As a whole, 3D printers facilitate manufacturing by:

- Helping re-engineer lighter and better designed versions of existing parts

- Being cost-effective for design changes
- Allowing low-risk manufacture for market testing
- Promoting customisation and personalised products
- Improving accuracy
- Decreasing costly mistakes
- Promoting collaborative designs between departments
- Highlighting costly design flaws
- Improving the customer and designer involvement and, hence, overall satisfaction
- Speeding up decision-making

Growing trends in technology

3D printing technology is evolving through development in three areas: printers and printing methods,

3D printing technologies commonly available in the market

Fused deposition modelling (FDM). It is an extrusion based technology. A stream of melted thermoplastic material is extruded from a nozzle to create layers, with each layer bonding to the previous layer.

Stereo lithography (SLA). This technology is also known as opto-fabrication. It creates products in a layer by layer fashion by using ultraviolet (UV) light beam to harden liquid resin that bonds each successive layer.

Polyjet and multijet modelling (MJM) systems. Similar to SLA, these use UV light to crosslink photopolymer. However, rather than scanning a laser to bond layers, a printer jet sprays tiny droplets of the photopolymer in the shape of the first layer. Then UV light beam crosslinks the polymer and fixes the shape of the layer and creates the build platform. Then more material is deposited layer by layer.

Selective laser sintering (SLS). It is similar to FDM. But instead of the liquid polymers used in FDM, it uses powdered materials such as titanium, aluminium, polystyrene, nylon and glass. Powder is sprayed onto the print surface and then laser is used to sinter the powder, layer by layer.

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